SuprMentr – Assignment(9th Feb 2026):

Assignment: Internet Explorer.

Draw and explain the client-server architecture and trace what happens when you open a website.

Ans: **1. Client–Server Architecture**

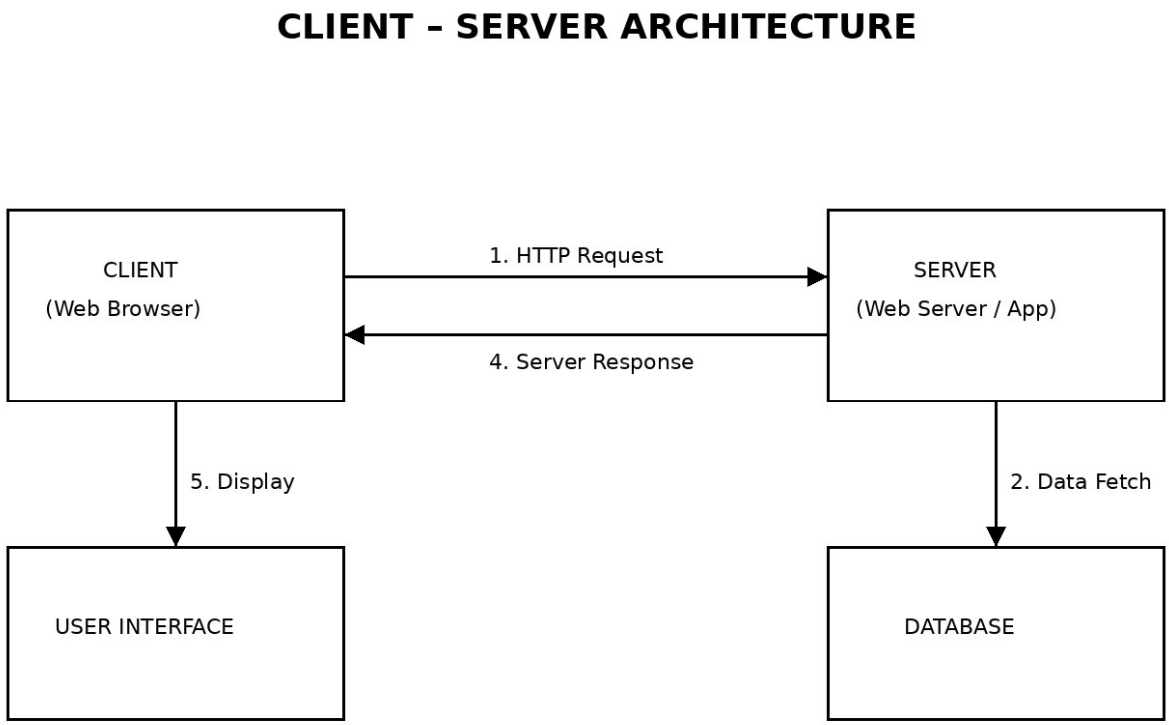

Fig: Client–Server Architecture.

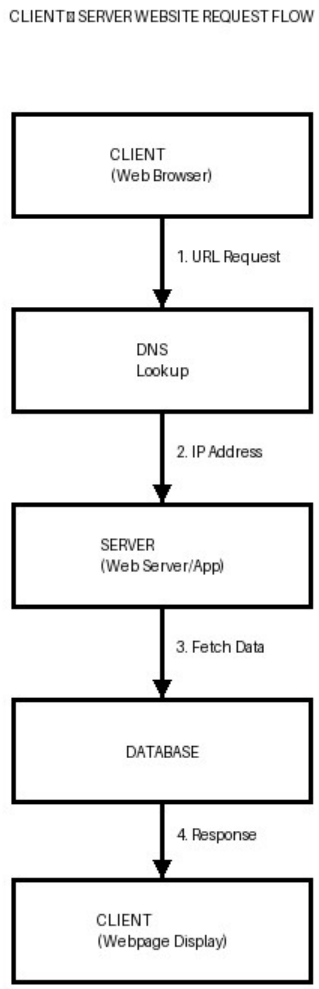

Fig: Website Request Flow.

Explanation:

Client–Server Architecture is the basic model on which the internet and web applications work.

- **Client**

  The client is the user's device or software that requests services.

  Examples:

  - Web browsers(Chrome, Edge, Firefox, Internet Explorer).
  - Mobile apps.

  Functions:

  - Sends requests to servers.
  - Displays webpage content.

    - Handles user interaction.

- **Server**

  The **server** is a powerful computer/system that provides resources or services.

  Examples:

    - Web servers(Apache, Nginx).
    - Application servers.
    - Cloud servers.

  Functions:

    - Processes client requests.
    - Runs backend code.
    - Retrieves data from database.
    - Sends response back.

- **Database(Supporting Component)**

    - Stores website data.
    - Example: user accounts, posts, products.

Simple Real-Life Example:

Client = You ordering food on Swiggy.
Server = Swiggy system processing order.
Database = Restaurant menu + user details.

2. What Happens When You Open a Website?

Example: www.google.com

Step-by-Step Process

- **User Enters URL**

  You type:

    www.google.com in the browser address bar.

  Browser = Client.

- **DNS Lookup Happens**

    - Browser asks: "Where is this website located?"
    - DNS(Domain Name System) converts domain → IP address.

Example:

www.google.com → 142.250.xxx.xxx

Like converting contact name → phone number.

- **Browser Sends HTTP/HTTPS Request**

  Browser sends a request to server:

  GET / HTTP/HTTPS

  This request travels through the internet.

- **Request Reaches Web Server**

  Server receives request and checks:

  - What page is requested?
  - Any backend processing needed?

- **Server Processes the Request**

  Server may:

  - Run backend code(Node.js, Python, Java, etc).
  - Fetch data from database.
  - Authenticate user(if login).

- **Server Sends Response**

  Server returns:

  - HTML → Structure.
  - CSS → Styling.
  - JavaScript → Functionality.
  - Images / Videos.

  This is called **HTTP Response**.

- **Browser Renders the Webpage**

  Browser:

  - Reads HTML.
  - Applies CSS.
  - Executes JavaScript.
  - Displays final webpage visually.

  Now you see Google homepage.

Process Flow:

1. Enter URL.
2. DNS converts domain → IP.
3. Browser sends request.
4. Server receives request.
5. Server processes data.
6. Server sends response.
7. Browser displays webpage.

3. Components Involved

Component	Role
Browser	Sends request & displays page
DNS	Converts domain to IP
Internet	Transfers data
Server	Processes request
Database	Stores website data

4. Importance of Client–Server Architecture

- Enables web browsing.
- Supports web applications.
- Handles multiple users.
- Centralized data storage.
- Improves scalability & security.

5. Conclusion

The client–server architecture forms the backbone of modern web development. It enables communication between users and web services through structured request and response cycles. Understanding how a

browser interacts with servers, DNS and databases helps in building efficient full stack applications. This knowledge provides a strong conceptual foundation for learning frontend, backend and API communication in future development tasks.